JACKSON SHEPPARD

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PH.D Candidate IN CHEMISTRY

SUMMARY

Ph.D candidate in the Department of Chemistry and Biochemistry at the University of California, Santa Barbara with a B.S in physics (Honors) from the same institution. Research experience spans molecular simulation, machine learning-driven molecular design, and quantitative analysis of complex biophysical systems. Previously worked as a controls engineer at SLAC National Accelerator Laboratory in a collaborative software development environment, earning the SLAC Spot Award for Dependability. Strong background in Python-based data analysis, database technologies, and scientific computing, with demonstrated excellence in written and oral communication.

TECHNICAL SKILLS

Molecular Simulation & Biophysics:

- **Molecular Dynamics:** GROMACS, AMBER, PLUMED, Replica Exchange MD (REMD), Monte Carlo
- **Analysis & Visualization:** MDAnalysis, VMD, custom Python workflows
- **Docking & Screening:** AutoDock Vina
- **Simulation Environments:** Linux, HPC workflows, CMake

Programming & Scientific Computing:

- **Languages:** Python, C++, R, MATLAB, SQL
- **Python Ecosystem:** NumPy, pandas, Matplotlib, scikit-learn, TensorFlow/Keras
- **Data & ML:** Machine learning, neural networks, NLP, ETL pipelines

Data Systems & Software Engineering:

- **Databases:** PostgreSQL, MongoDB, SQLite
- **Web & Visualization:** Flask, Plotly, D3.js, Leaflet, GeoJSON, Bootstrap
- **Cloud & Tools:** AWS (S3, RDS), Jupyter, Google Colab
- **Version Control:** Git/GitHub, SVN

Controls & Instrumentation

- **Control Systems:** EPICS, PLC programming, Beckhoff
- **Applications:** EPICS Display Manager, Tc2_MC2

PROJECTS

Hydraulic Activation of the AsLOV2 Photoreceptor | https://doi.org/10.1101/2025.06.19.660617

Demonstrated that blue-light activation of the LOV2 (light–oxygen–voltage) domain of Avena sativa phototropin 1 (AsLOV2) induces concerted water motion that drives protein conformational extension, and that this activation can be triggered by either blue light or high pressure..

- Submitted 2025
- Role: Contributing Author
- Tools: GROMACS, Amber, PLUMED SASA module, MDAnalysis, Python

Pareto-Optimal Molecular Discovery with Thompson-Sampled Scaffold Allocation | In Progress

Developed a reinforcement learning–driven framework to inhibit α-synuclein aggregation in aging-associated neurodegeneration, integrating multi-objective optimization and molecular simulation for compound validation.

- Role: Contributing Author
- Tools: GROMACS, AutoDock Vina QuickVina2-GPU, PyTorch, RDKit, MDAnalysis, Python

Biasing Chemical Language Models Using Molecular Simulation | In Progress
Used molecular dynamics simulations of protein–ligand complexes to identify interaction motifs that modulate protein stability and conformational change, and leveraged these insights to guide chemical language models toward desired protein conformations.

- Role: Contributing author
- Tools: GROMACS, MDAnalysis, Python

Experience

Ph.D Student August 2021 - Present
University of California, Santa Barbara Santa Barbara, CA
Ph.D candidate in the Shea Group within the Department of Chemistry and Biochemistry at UCSB. Primary project involves the development of an end-to-end computational framework for rational drug design using large language models optimized using a multi-objective reinforcement learning loop. Secondary project characterizes the relationship between protein structure and interfacial water, quantifying the change in distribution of low-entropy, tetrahedrally coordinated "wrap" water that drives long-range conformational changes.
Key Accomplishments:

- Developed computational framework for small molecule discovery in collaboration with the Ambuj Singh group (UCSB Computer Science), incorporating large language models fine-tuned on chemical datasets, reinforcement learning based optimization via protein-ligand and drug-like metrics, and extensive replica exchange molecular dynamics simulations of top-scoring molecules.
- Developed Python code to measure the three-body angle distribution of water molecules over explicit solvent molecular simulations in collaboration with the Songi Han (Northwestern Chemistry) and Mark Sherwin (UCSB Physics) groups. Calculated population distributions and residence times of low-density tetrahedral-oriented water and high-density icosahedral-oriented water.
- Mentored undergraduate students with a focus on developing computational problem solving skills and working in a collaborative scientific environment.

Controls Support November 2020 - July 2021
SLAC National Accelerator Laboratory Menlo Park, CA
Part-time position in the Experiment Control Systems Delivery (ECS, formerly PCDS) group within the Linac Coherent Light Source (LCLS) while attending the UC Berkeley Extension Data Analytics Boot Camp. Responsible for remote support of LCLS-II style Beckhoff motion control systems.
Key Accomplishments:

- Created procedures for setup, basic operation, and troubleshooting of motion control systems under expertise.
- Coordinated installation and checkout of motion controls for CVMI interaction point at the TMO endstation.
- Trained recent hires and colleagues on systems under expertise to create a more uniform understanding within the group.

Science and Engineering Associate September 2018 - November 2020
SLAC National Accelerator Laboratory Menlo Park, CA
Controls engineer in the Experiment Control Systems (ECS) Delivery group within the Linac Coherent Light Source (LCLS). During active operations, responsible for experiment setup and on-call technical support of assigned experiments at assigned instruments. Over instrument downtime, responsible for integration of new LCLS-II devices into ECS-developed control systems.
Key Accomplishments:

- Integrated motors, cameras, temperature sensors, timing trigger signals, and other user devices into the ECS-developed controls software stack for assigned experiments.
- Expanded device support through Python development on a Linux system (IPython sessions containing device objects for specific experiments).
- Deployed a controls upgrade to existing Offset Mirror System to employ LCLS-II style Beckhoff motion hardware and PLC software interface. Coordinated system reinstallation and performed checkout to ensure motion requirements were satisfied. Integrated system into ECS controls stack for use by scientists during experiments.
- Received SLAC Spot Award for Dependability (August 4, 2020).

Summer Student June 2018 - September 2018
SLAC National Accelerator Laboratory Menlo Park, CA
Student under the mentorship of staff physicist Claudio Pellegrini. Worked on free electron laser (FEL) beam dynamics simulations using GENESIS and MATLAB for preprocessing and analysis.
Key Accomplishments:
- Characterized the undulator "taper profile" by finding the optimal relationship for magnetic field strength as a function of longitudinal distance using simulations and iterative search methods.
- Presented findings in the LCLS Summer Internship Poster Session with other students, mentors, and SLAC faculty (Received LCLS Poster Award, 2nd place).
- Published findings in the Journal of Synchrotron Radiation: Halavanau, A., Decker, F. J., Emma, C., Sheppard, J., Pellegrini, C. 2019. Very high brightness and power LCLS-II hard X-ray pulses. *J. Synchrotron Rad.* **26(3)**:635-646.

Undergraduate Research Assistant January 2018 - June 2018
University of California, Santa Barbara Santa Barbara, CA
Student in the Shea Group within the Department of Chemistry at UCSB. Participated in computational research that applied statistical and data science techniques to molecular dynamics simulations in order to model complex biological processes.
Key Accomplishments:
- Cleaned and pre-processed data using NumPy and employed cluster analysis techniques using Python to prepare data for models built with TensorFlow and Keras.
- Produced latent space visualization with Matplotlib.
- Utilized 3D modeling software Visual Molecular Dynamics to visualize protein folding.
- Earned Department of Physics Academic Honors (May 13, 2018).

Education

Ph.D, Chemistry: University of California, Santa Barbara, Santa Barbara, CA
Expected 2027

Data Analytics Certificate: UC Berkeley Extension, Berkeley, CA
May 2021
A 24-week intensive program focused on gaining technical programming skills in Excel, VBA, Python, R, JavaScript, SQL Databases, Tableau, Big Data, and Machine Learning.

Bachelor of Science, Physics: University of California, Santa Barbara, Santa Barbara, CA
June 2018
GPA: 3.79 (High Honors)